

HTTPS, SOAP/REST, FTP, Database via JDBC, LDAP, message-oriented middleware (MOM) via JMS, Mail - SMTP(S), POP3(S) and IMAP(S), MongoDB (NoSQL), Native commands/shell scripts, TCP.

- A full multi-threading framework allows concurrent sampling by many threads and simultaneous sampling of different functions by separate thread groups.
- A well-planned GUI design allows faster test plan building and debugging.
- Caching and offline analysis/ replaying of test results.
- Plugin support for data analysis and visualisation; pluggable samplers allow unlimited testing capabilities. Apache JMeter can be used to test performance on both static and dynamic resources (Web services - SOAP/ REST; Web dynamic languages - PHP, Java, ASP.NET, files, etc; Java objects, databases and queries, FTP servers and more). It can be used to simulate a heavy load on a server, a group of servers, network or object to test its strength, or to analyse overall performance under different load types. You can use it to make a graphical analysis of performance or to test the behaviour of your server, script and object under a heavy concurrent load.

Selenium



Figure 2: Selenium Web automation

Selenium is one of the most popular open source testing tools for Web applications. It was originally developed by Jason Huggins in 2004 for testing an internal

project at ThoughtWorks.

Selenium supports automation testing of Web based applications for some of the largest browser vendors. It has five basic components — the Selenium IDE, the Selenium Remote Control, the Selenium WebDriver, the Selenium Grid and Selenium Client API (introduced in Selenium 2). The

Selenium IDE is a Firefox add-on for record-and-playback Web application tests. WebDriver directly communicates with the Web browser and uses its native compatibility to automate functions.

OpenSTA or Open Systems Testing Architecture



Figure 3: OpenSTA - Open Systems Testing Architecture

OpenSTA is a GUI-based Web server benchmarking utility. The

current toolset has the capability to perform scripted HTTP and HTTPS heavy load tests with performance measurements from Win32 platforms. OpenSTA collects results and statistics during test runs by a variety of automatic and user controlled mechanisms. These include scripted timers, SNMP data, Windows Performance Monitor stats and HTTP results and timings. Much of the data logged can be monitored live during the test runs; once test runs are complete, logs can be viewed, graphed, filtered and exported for use by more sophisticated report generation software.

The Grinder



Figure 4: The Grinder

The Grinder is a Java load testing framework that makes it easy to run a distributed test using many load injector machines. The Grinder was originally developed by Paco Gómez and Peter Zadrozny.

Its key features are:

- Can load test anything that has a Java API. This includes common cases such as HTTP Web servers, SOAP and REST Web services, and application servers (CORBA, RMI, JMS, EJBs), as well as custom protocols.
- Supports powerful scripting languages like Jython and Clojure.
- It has a distributed framework, with a user-friendly graphical console that allows multiple load injectors to be monitored and controlled, and provides centralised script editing and distribution.

- Mature HTTP support, automatic management of client connections and cookies, SSL, is proxy aware, has connection throttling, a sophisticated record-and-replay of the interactions between a browser and a website, etc.

Vega



Figure 5: Vega - A security testing tool

Vega is a free open source website security testing tool. It is GUI based, written in Java, and runs on Windows, Linux and OS X. Vega can help you find and validate SQL injection, cross-site scripting (XSS), inadvertently disclosed sensitive information, and other vulnerabilities.

Vega is developed by Subgraph, an open source security startup based in Montreal, Canada. It supports automated scanning and has an intercepting proxy testing mode. In automated scanner mode, it automatically crawls websites, extracting links, processing forms, and running modules for possible injection points it discovers. These modules can do things like automatically submitting requests that fuzz parameters, for example, to test for things like cross-site scripting (XSS) or SQL injection.

The intercepting proxy mode allows you to test Web applications on the proxy server. When a browser uses the Vega proxy, requests and responses are visible to Vega. It can be instructed to set 'breakpoints', interception criteria for outgoing requests (from the browser) or incoming responses (from the server).

SiteDigger

SiteDigger is an open source tool to examine Google's cache for errors, vulnerabilities, configuration issues, proprietary information and interesting security nuggets on websites.

SiteDigger runs on a Windows machine and it requires the Microsoft .NET Framework v3.5. SiteDigger doesn't require the Google API License Key. Its features include proxy support, real-time results, configurable result sets, etc.